# Watch-DevKit-HW

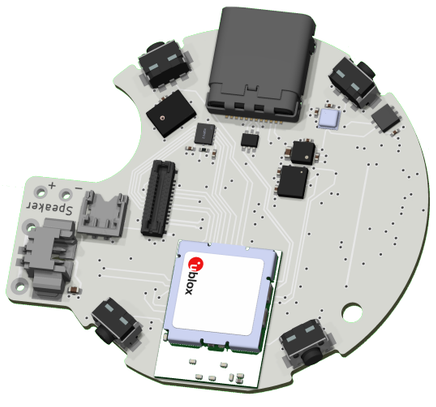
## Variant: PRELIMINARY

# 2026-10-21

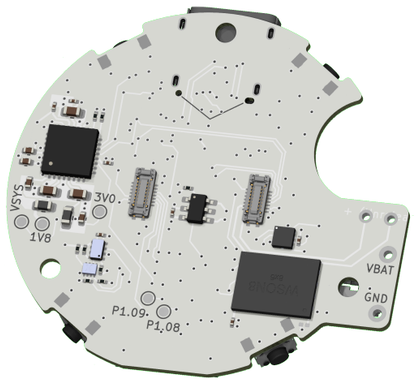
# Rev + (Unreleased)


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## TOP VIEW




## BOTTOM VIEW




## NOTES

Comment

Not fitted components are marked as X

DRAFT - Very early stage of schematic, ignore details.
PRELIMINARY - Close to final schematic.
CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.
RELEASED - A board with this schematic has been sent to production.

Date: 21-Oct-2026

## DESIGN CONSIDERATIONS

DESIGN NOTE: Example text for informational design notes.

DESIGN NOTE: Example text for debug notes.

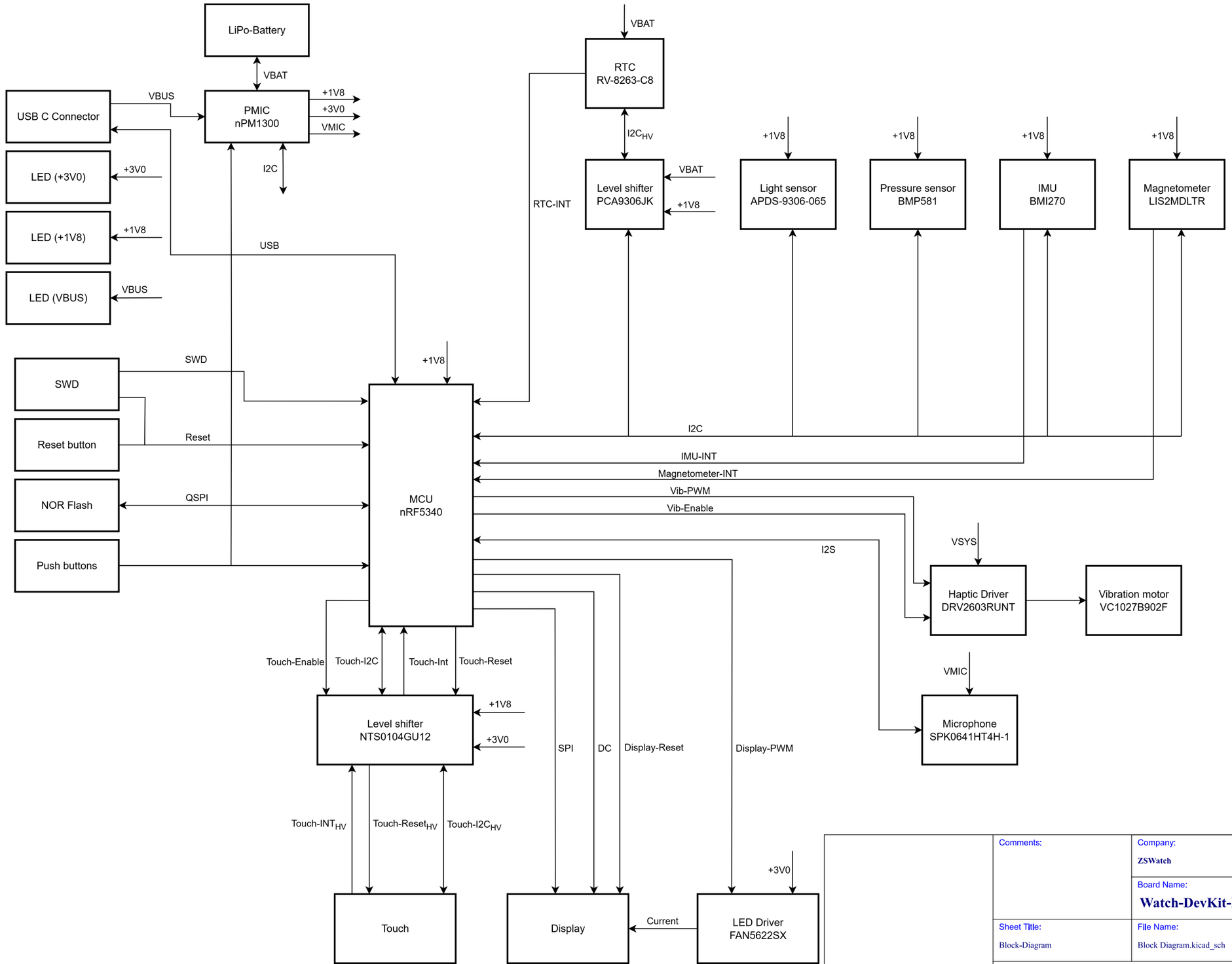
DESIGN NOTE: Example text for cautionary design notes.

**DESIGN NOTE: Example text for critical design notes.**

**LAYOUT NOTE: Example text for critical layout guidelines.**

| | | | | |
|---|---|---|---|---|
| Comments: | Company: ZSWatch | | Variant: PRELIMINARY | Git Hash: fccc54e |
| | Board Name: Watch-DevKit-HW | | Project Name: ZSWatch Watch-DevKit | |
| Sheet Title: ZSWatch-Watch | File Name: ZSWatch-Watch.kicad_sch | Designer: Daniel Kampert | Date: 2025-07-13 | Revision: + (Unreleased) |
| Sheet Path: / | | Reviewer: | Size: A3 | Sheet: 1 of 9 |

# [2] Block-Diagram




## Target specifications:

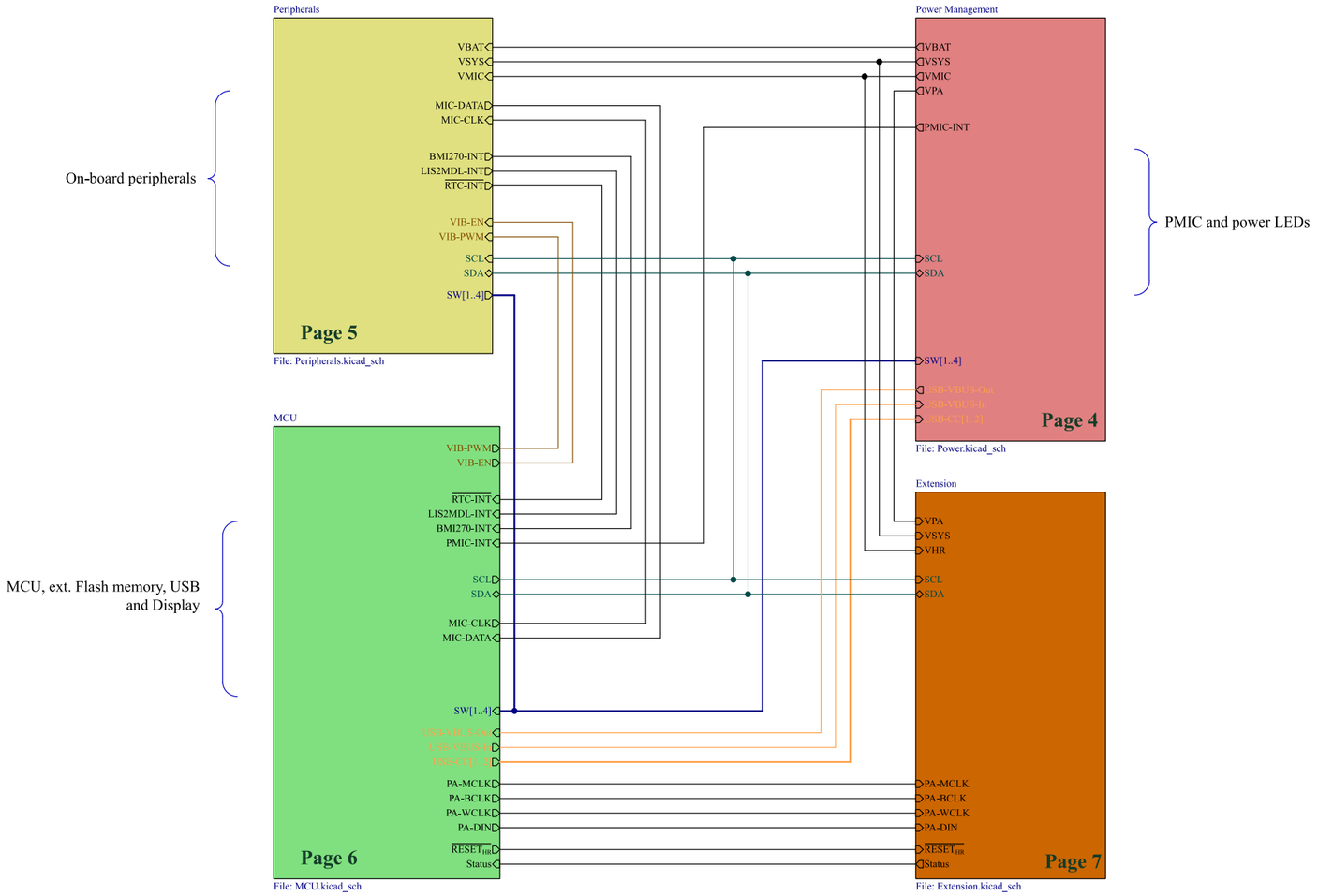
| | |
|---|---|
| **Input voltage:** | 5 V ± 5% |
| **Battery voltage** | 3.2 V to 4.2 V |
| **Spec 3** | ?? |
| **Spec 4** | ?? |

| | | | |
|---|---|---|---|
| Comments: | Company: ZSWatch | Variant: PRELIMINARY | Git Hash: fccc54e |
| | Board Name: Watch-DevKit-HW | Project Name: ZSWatch Watch-DevKit | |
| Sheet Title: Block-Diagram | File Name: Block Diagram.kicad_sch | Designer: Daniel Kampert | Date: 2025-07-13 / Revision: + (Unreleased) |
| Sheet Path: /Block Diagram/ | | Reviewer: | Size: A3 / Sheet: 2 of 9 |

**Target specifications:**

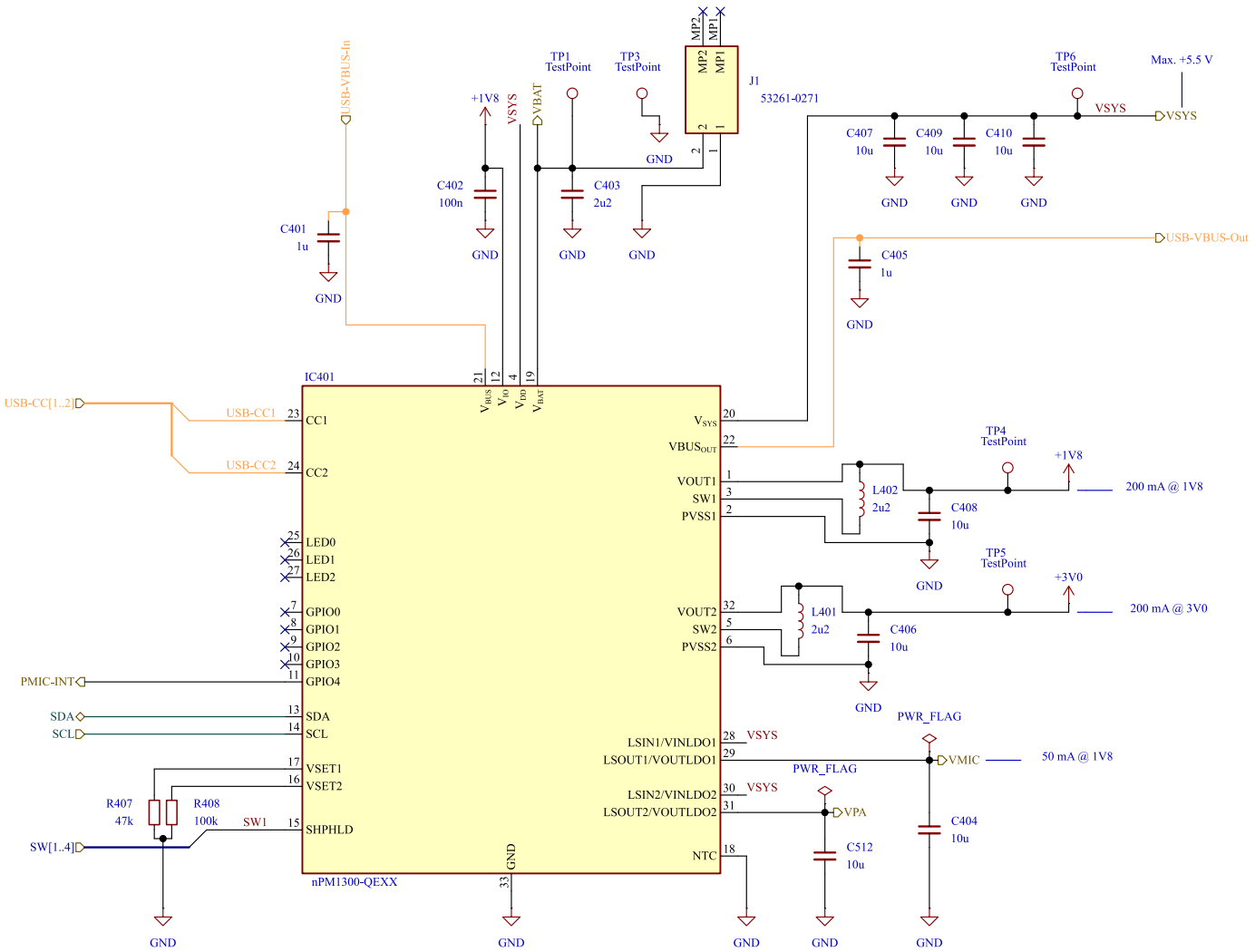
| | |
|---|---|
| **Input voltage:** | ?? - ?? V |
| **Spec 2** | ?? |
| **Spec 3** | ?? |
| **Spec 4** | ?? |

| | | | | |
|---|---|---|---|---|
| Comments: | Company: **ZSWatch** | | Variant: PRELIMINARY | Git Hash: fccc54e |
| | Board Name: **Watch-DevKit-HW** | | Project Name: **ZSWatch Watch-DevKit** | |
| Sheet Title: | File Name: Power - Topology.kicad_sch | Designer: Daniel Kampert | Date: 2025-07-13 | Revision: + (Unreleased) |
| | Sheet Path: /Power - Topology/ | Reviewer: | Size: **A3** | Sheet: **2** of **9** |




| I2C addresses | |
|---|---|
| BMP581 | 0x46 |
| RV-8263-C8 | 0x51 |
| APDS-9306-065 | 0x52 |
| BMI270 | 0x68 |
| nPM1300 | 0x6B |
| LIS2MDLTR | 0x1E |

| | | | | |
|---|---|---|---|---|
| Comments: | Company: ZSWatch | | Variant: PRELIMINARY | Git Hash: fccc54e |
| | Board Name: **Watch-DevKit-HW** | | Project Name: **ZSWatch Watch-DevKit** | |
| Sheet Title: Power-Management | File Name: Project Architecture.kicad_sch | Designer: Daniel Kampert | Date: 2025-07-13 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/ | | Reviewer: | Size: **A3** | Sheet: **3** of **9** |

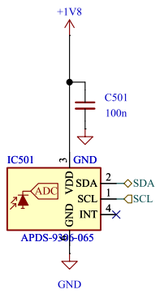



PWR_FLAG

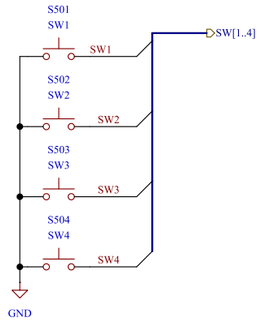
GND

| | | | | |
|---|---|---|---|---|
| Comments: | Company: ZSWatch | | Variant: PRELIMINARY | Git Hash: fccc54e |
| | Board Name: **Watch-DevKit-HW** | | Project Name: **ZSWatch Watch-DevKit** | |
| Sheet Title: Power-Management | File Name: Power.kicad_sch | Designer: Daniel Kampert | Date: 2025-07-13 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Power Management/ | | Reviewer: | Size: **A4** | Sheet: **4** of **9** |

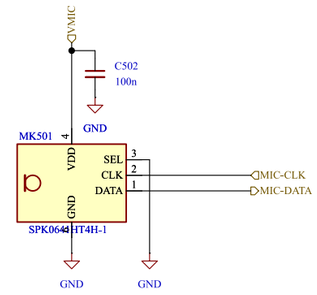
## Light sensor



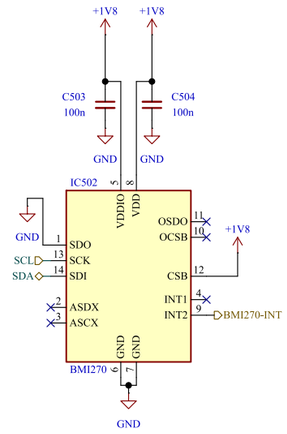

## Buttons



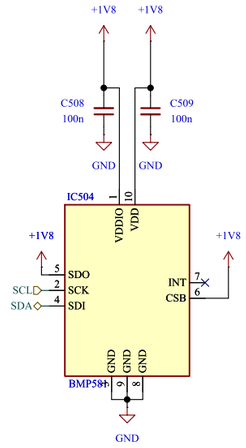

## Microphone



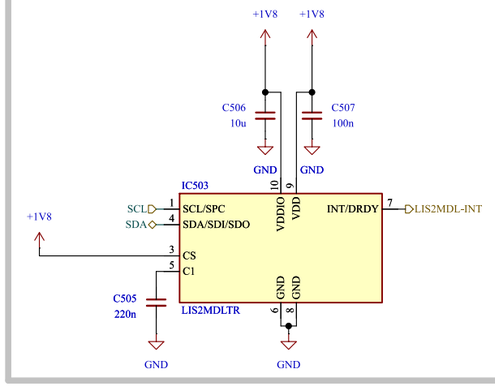

## IMU with step counter and gesture detection



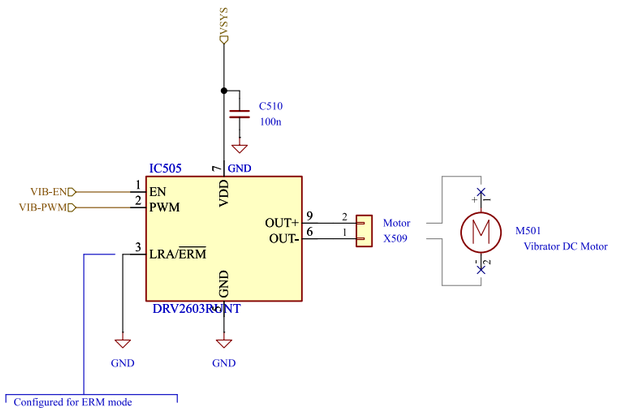

## High accuracy pressure sensor



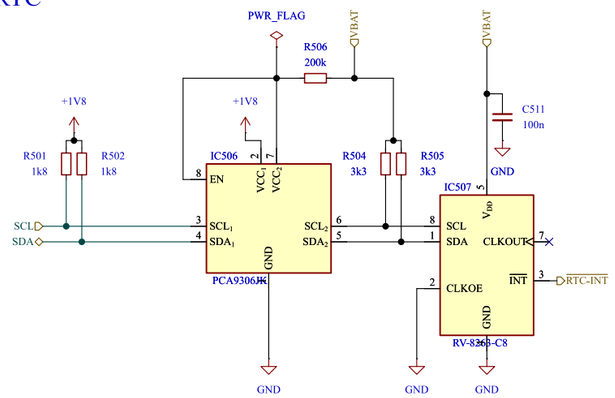

## Magnetometer




## Vibration motor

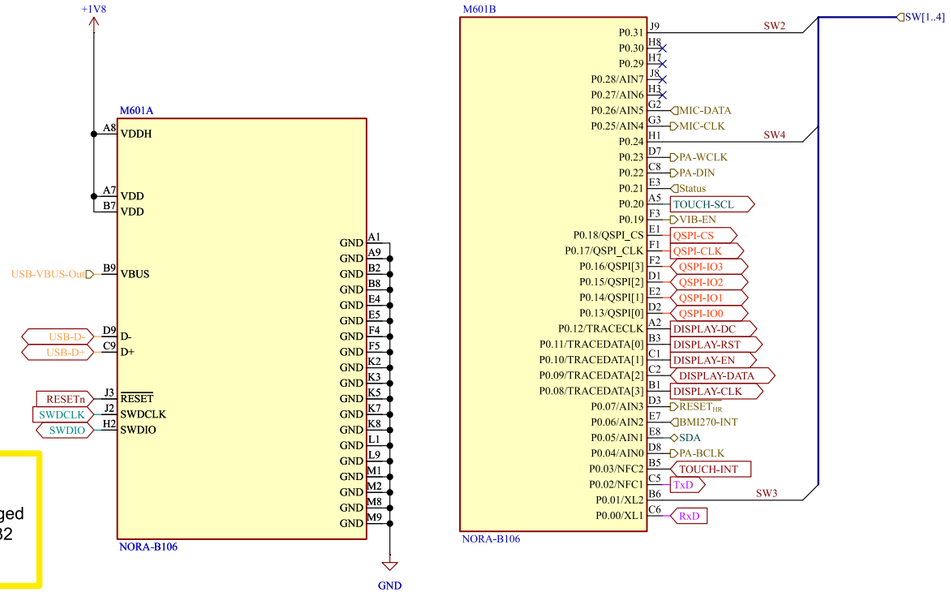



## RTC



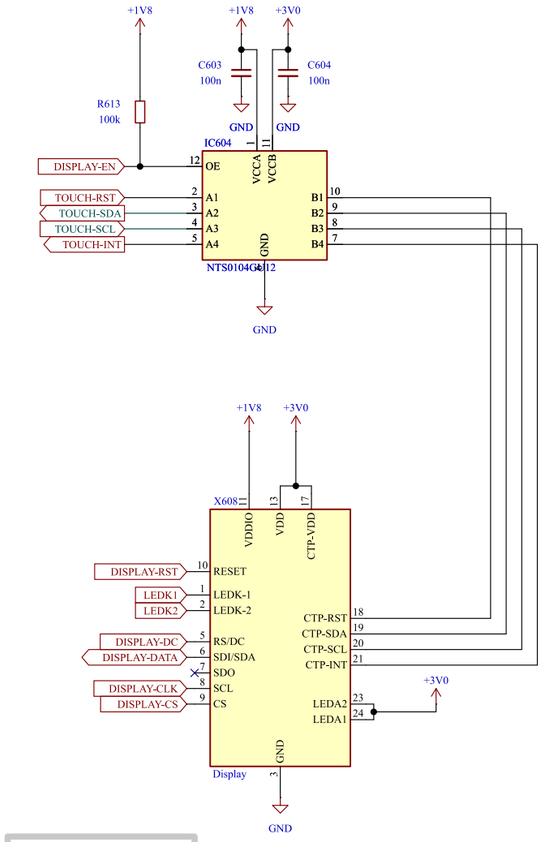

| | | | | |
|---|---|---|---|---|
| Comments: | Company: **ZSWatch** | | Variant: PRELIMINARY | Git Hash: fccc54e |
| | Board Name: **Watch-DevKit-HW** | | Project Name: **ZSWatch Watch-DevKit** | |
| Sheet Title: Peripherals | File Name: Peripherals.kicad_sch | Designer: Daniel Kampert | Date: 2025-07-13 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/Peripherals/ | | Reviewer: | Size: **A3** | Sheet: **5** of **9** |

## Microcontroller (NORA-B106/nRF5340)



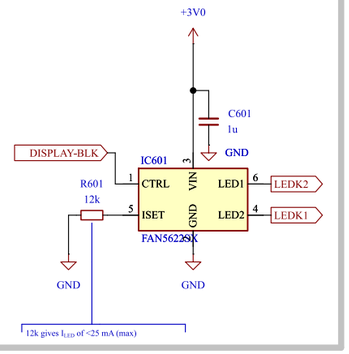

DESIGN NOTE:
The SPI lines for the display can´t be changed and are required for 32 MHz operation.

## Display

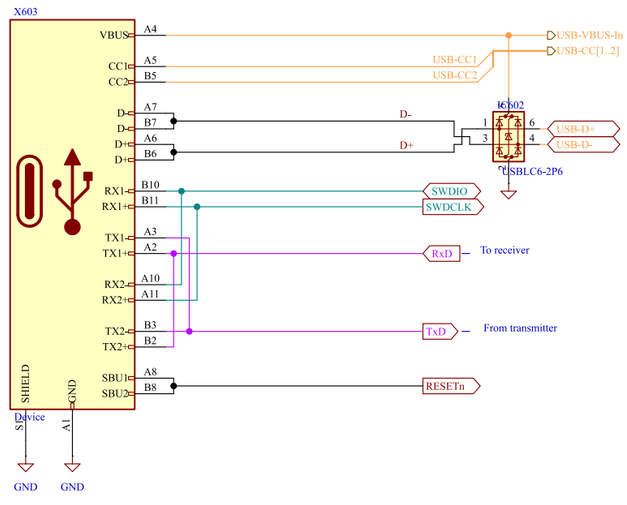



DESIGN NOTE:
NTS0104GU12 has internal pullups.

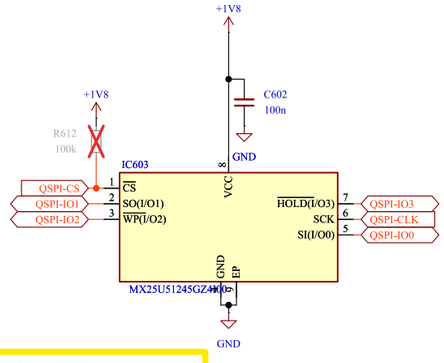
## Display backlight




## USB

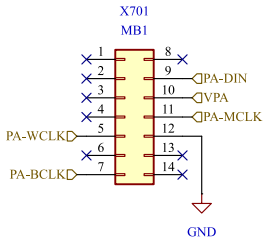


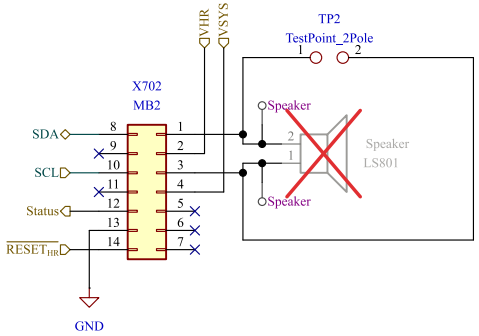

## External Flash




DESIGN NOTE:
Make sure that the Flash Memory supports
4k sector erase commands (limited by QSPI driver)!

| | | | | |
|---|---|---|---|---|
| Comments: | Company: **ZSWatch** | | Variant: PRELIMINARY | Git Hash: fccc54e |
| | Board Name: **Watch-DevKit-HW** | | Project Name: **ZSWatch Watch-DevKit** | |
| Sheet Title: MCU | File Name: MCU.kicad_sch | Designer: Daniel Kampert | Date: 2025-07-13 | Revision: + (Unreleased) |
| Sheet Path: /Project Architecture/MCU/ | | Reviewer: | Size: **A3** | Sheet: **6** of **9** |




| | | | | |
|---|---|---|---|---|
| | Comments: | Company: **ZSWatch** | Variant: PRELIMINARY | Git Hash: fccc54e |
| | | Board Name: **Watch-DevKit-HW** | Project Name: **ZSWatch Watch-DevKit** | |
| | Sheet Title: Extension | File Name: Extension.kicad_sch | Designer: Daniel Kampert | Date: 2025-07-13 / Revision: + (Unreleased) |
| | Sheet Path: /Project Architecture/Extension/ | | Reviewer: | Size: **A4** / Sheet: **7** of **9** |

# [8] Revision-History

**Version 1.0.0 - 2026-02-21**

**Added:**

Initial release

**Version 1.1.0 - 2026-10-16**

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Added

- Test pads for FEM pins, so NORA-W126 (drop-in compatible, with FEM for higher output power) can be mounted instead of NORA-B106 (#21)

- Pads for speaker instead of a speaker connector (#18)

- Extra power test points on the bottom side of the board (#18)

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Changed

- Larger 1.25 mm pitch battery connector (Molex 53261-0271) (#18)

- Battery and speaker stickout reduced by 0.5 mm (#18)

- Optional speaker LS801 excluded from the board and BOM (#18)

| | | | | |
|---|---|---|---|---|
| Comments: | Company: ZSWatch | | Variant: PRELIMINARY | Git Hash: fccc54e |
| | Board Name: Watch-DevKit-HW | | Project Name: ZSWatch Watch-DevKit | |
| Sheet Title: Revision-History | File Name: Revision History.kicad_sch | Designer: Daniel Kampert | Date: 2025-07-13 | Revision: + (Unreleased) |
| Sheet Path: /Revision History/ | | Reviewer: | Size: A4 | Sheet: 8 of 9 |